AKTU

B.Tech I-Year

Electrical Engg.

ONE SHOT Revision

Unit-3

Transformers

Avinash Sir

GW GATEWAY CLASSES

Gateway Classes

- **Unit-3: (Transformers)**
- Magnetic circuits,
- ideal and practical transformer, → 1
- equivalent circuit, V.Imp 1 Question
- losses in transformers,
- regulation and efficiency. → 1 Ques

Gateway classes

Basic Definitions

1. **Magnetic Field:-** Area around a magnet within which it influence its perceptible is known as magnetic field

2. **Magnetic Flux :-** total number of magnetic field lines passing through a given surface. unit of flux is Weber (Wb). 1 wb = 10^8 Lines line of force

3. **Magnetic Flux density :-** Flux per unit area is known as flux density $(B) = \frac{flux}{area}$

4. **Magnetic field intensity:-** Magnetizing force or Magnetic field intensity or magnetic field strength is the MMF required to magnetize a unit length of the magnetic flux path. The unit of magnetic field intensity is AT/m and is denoted by H.

$$H = \frac{NI}{l} \qquad \text{AT/m}$$

Gateway classes

Basic Definitions

5. Magneto-motive force or MMF:- MMF is the cause for producing the magnetic flux. The MMF in a magnetic circuit depends on the number of turns(N) and the amount of current(I) flowing through it.

$\boxed{MMF = NI}$ AT unit is ampere turns.

(6)Reluctance:- It is the opposition that the magnetic circuit offers for the flow of magnetic flux. We can also define the reluctance as the ratio of magneto-motive force to the magnetic flux. It is denoted by S and its unit is ampere-turns per weber.

$$Reluctance(S) = \frac{mmf}{Flux} = \frac{l}{A\mu_o\mu_r} = \frac{l}{\mu A}$$

$\rho \rightarrow \frac{1}{\mu_o \mu_r}$ $\frac{Emf}{I} = \frac{V}{I} = R = \rho \frac{l}{a}$

7. Permeance:- Permeance is the reciprocal of reluctance. The ease with which the flux can pass through the material is known as permeance. Weber/AT is the unit of permeance.

8. Permeability:- Permeability is the ability of a material to allow the magnetic flux when the object is placed inside the magnetic field.

$$\mu = \mu_o\mu_r$$

Where μ_o is absolute permeability ($4\pi \times 10^{-7}$) and $\mu_r = 1\ for\ air$ is relative permeability

Gateway classes

Magnetic Circuit

Magnetic Circuit:- The closed path followed by the magnetic flux is magnetic circuit.

Ex →

The laws of magnetic circuits are quite similar to the electric circuits.

A **magnetic circuit** is made up of magnetic materials having high permeability such as iron, soft steel, etc. Consider a solenoid having N turns wound on an iron core. The magnetic flux of ø Weber sets up in the core when the current of I ampere is passed through a solenoid.

Let, l = mean length of the magnetic circuit, A = cross-sectional area of the core

μr = relative permeability of the core

Now the Field Strength is given by

$$H = \frac{NI}{l} \quad \text{AT/m}$$

$$B = \mu_o \mu_r H = \frac{\mu_o \mu_r NI}{l} \; Wb/m^2 \; \text{Wb}$$

$$\text{Total Flux} \emptyset = B \times A = \frac{\mu_o \mu_r NIA}{l} = \frac{NI}{\left(\frac{l}{A\mu_o\mu_r}\right)} \; \text{Wb.}$$

→ mmf

→ S

The denominator $\frac{l}{A\mu_o\mu_r}$ is called reluctance of the circuit

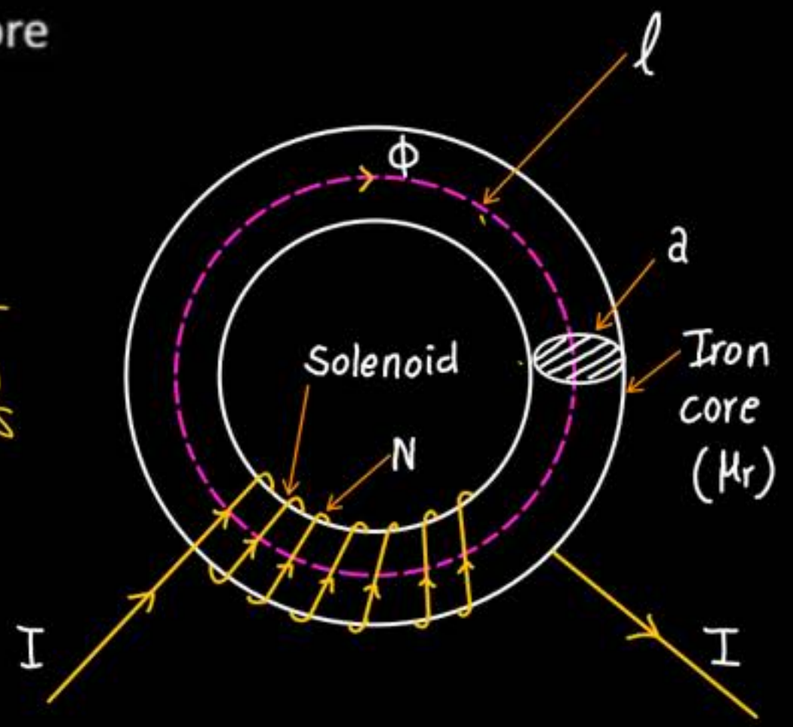
and NI is called mmf .

$$\emptyset = \frac{mmf}{reluctance} \text{Wb}$$

this relation is similar to electric circuit I = V/R

$$I = \frac{V}{R}$$

$$\phi = \frac{mmf}{?}$$



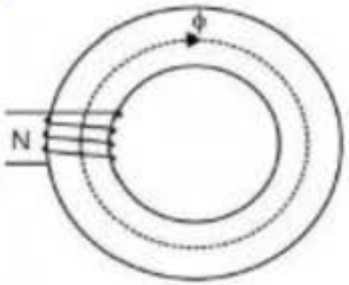
Gateway classes

Similarities between Magnetic Circuit & Electric Circuit

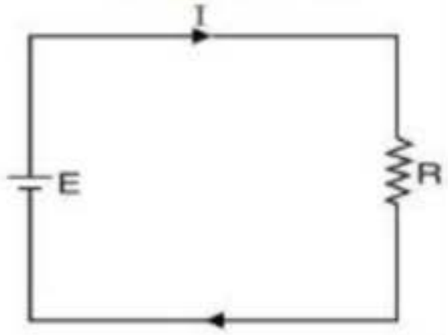
Similarities

1. The closed path for magnetic flux is called a magnetic circuit.	1. The closed path for electric current is called an electric circuit.
flux ϕ = mmf/Reluctance 2. Flux, $\phi = \dfrac{\text{m.m.f.}}{\text{reluctance}}$	$I = \dfrac{V}{R}$ 2. Current, $I = \dfrac{\text{e.m.f.}}{\text{resistance}}$
3. m.m.f. (ampere-turns) mmf	3. e.m.f. (volts) emf
4. Reluctance, $S = \dfrac{l}{a\mu_0\mu_r}$ S	4. Resistance, $R = \rho\dfrac{l}{a}$ R
5. Flux density, $B = \dfrac{\phi}{a}\ \text{Wb/m}^2$ B	5. Current density, $J = \dfrac{I}{a}\ \text{A/m}^2$ J
6. m.m.f. drop $= \phi S$	6. Voltage drop $= I R$
7. Magnetic intensity, $H = N I/l$	7. Electric intensity, $E = V/d$
8. Permeance	8. Conductance.
9. Permeability	9. Conductivity

Magnetic Circuit



Electric Circuit



Gateway classes

Disimilarities between Magnetic Circuit & Electric Circuit

(5) Magnetic flux can pass through air

Current can not pass through air

Dissimilarities

1. Truly speaking, magnetic flux does not flow.	**1.** The electric current acutally flows in an electric circuit.
2. There is no magnetic insulator. For example, flux can be set up even in air (the best known magnetic insulator) with reasonable m.m.f.	**2.** There are a number of electric insulators. For instance, air is a very good insulator and current cannot pass through it.
3. The value of μ_r is not constant for a given magnetic material. It varies considerably with flux density (B) in the material. This implies that reluctance of a magnetic circuit is not constant rather it depends upon B.	**3.** The value of resistivity (ρ) varies very slightly with temperature. Therefore, the resistance of an electric circuit is practically constant. This salient feature calls for different approach to the solution of magnetic and electric circuits.
4. No energy is expended in a magnetic circuit. In other words, energy is required in creating the flux, and not in maintaining it.	**4.** When current flows through an electric circuit, energy is expended so long as the current flows. The expended energy is dissipated in the form of heat.

Gateway classes

Types

- **Series magnetic circuits :-** consider a circular ring made from different materials of lengths l_1, l_2 and l_3 ,cross-sectional areas a_1, a_2 and a_3 and relative permeability μ_{r1}, μ_{r2} and μ_{r3} respectively with a cut of length lg called air gap. The total reluctance is the arithmetic sum of individual reluctance as they are joined in series.

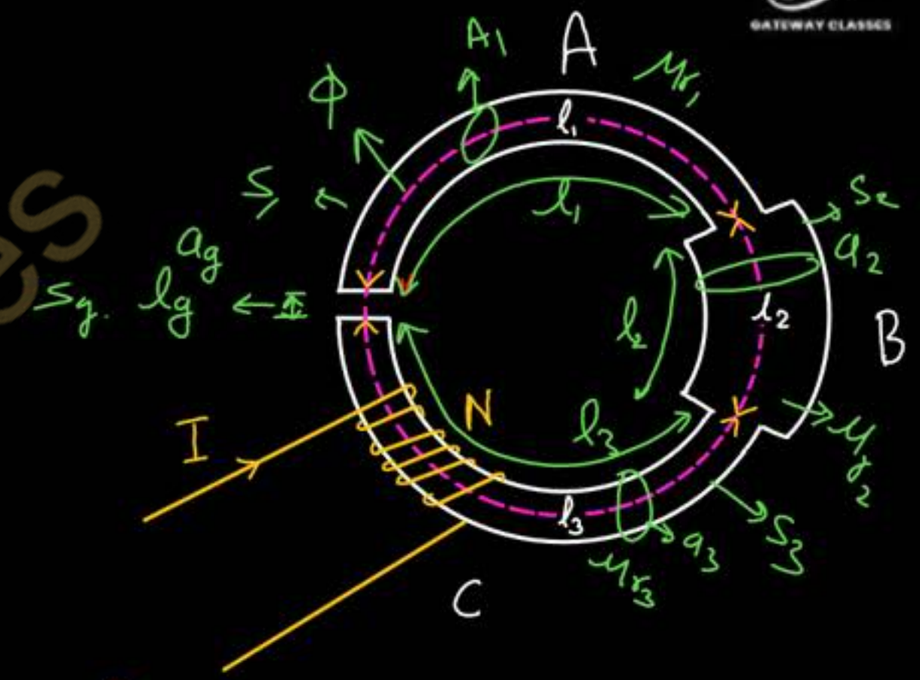
Total Reluctance $S = S_1 + S_2 + S_3 + S_g$

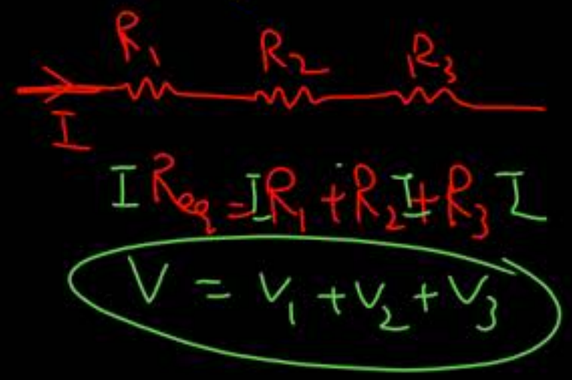
$$S = \frac{l_1}{\mu_0 \mu_{r_1} a_1} + \frac{l_2}{\mu_0 \mu_{r_2} a_2} + \frac{l_3}{\mu_0 \mu_{r_3} a_3} + \frac{l_g}{\mu_0 a_g}$$

Total mmf $= \phi \times S$

$$= \phi [S_1 + S_2 + S_3 + S_g]$$

$$\boxed{\text{mmf} = \phi S_1 + \phi S_2 + \phi S_3 + \phi S_g}$$



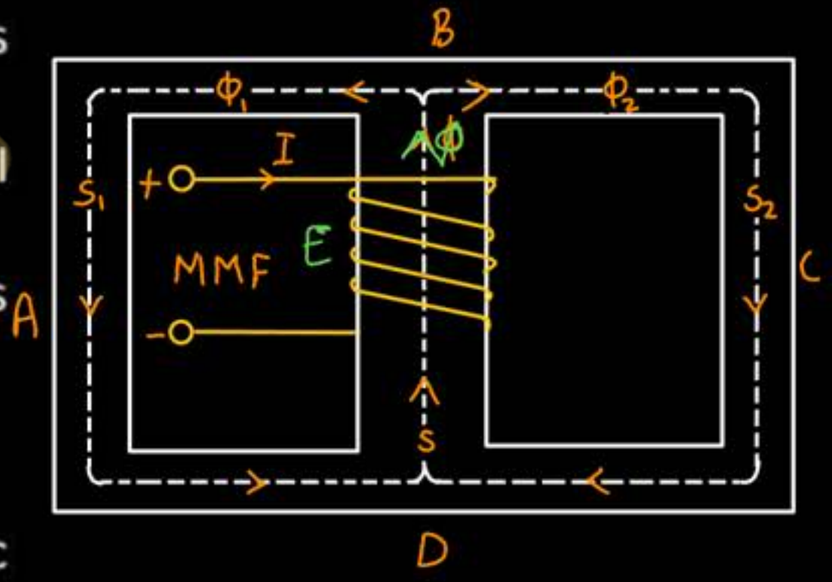


Gateway classes

A magnetic circuit which has more one path for the magnetic flux is called as ***parallel magnetic circuit.***

Consider a coil of N turns wound on limb BD carries an electric current of I amperes. The magnetic flux ϕ set up by the coil divides at B into two paths. The magnetic ϕ_2 passes along the path BCD. The magnetic flux ϕ_1 passes along the path BAD.

Therefore, the total flux is, $\Phi = \phi_1 + \phi_2$



The path BD and BCD are in parallel and hence form a parallel magnetic circuit. In a parallel magnetic circuit, the MMF required for the whole parallel magnetic circuit is equal to MMF required for any one of the parallel paths.

Let

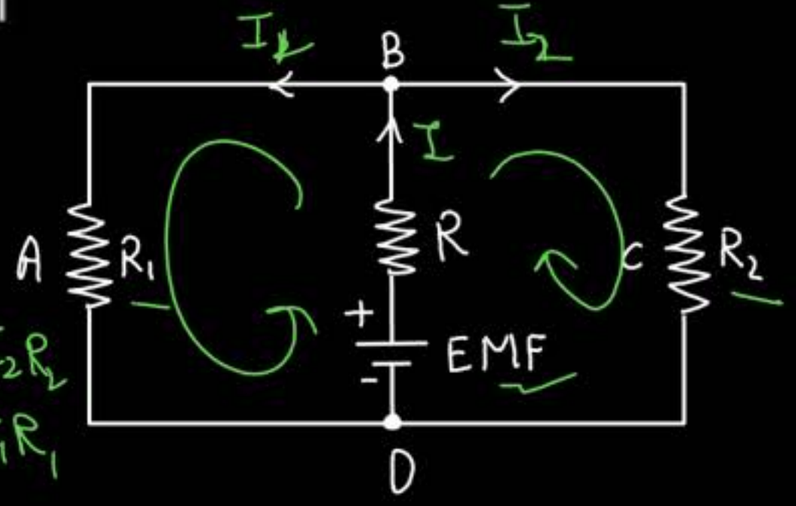
S = Reluctance of magnetic path BD, S_1 = Reluctance of magnetic path BAD

S_2 = Reluctance of magnetic path BCD

Therefore,

Total MMF=MMF for path ABD+MMF for path BD or BCD

$$MMF = \phi.S + \phi_2.S_2 = \phi.S + \phi_1.S_1$$



$= IR + I_2R_2$

$= IR + I_1R_1$

Gateway classes

Leakage Flux and Fringing

Leakage Flux:- The part of magnetic flux that does not follow the desired path in a magnetic circuit is known as *leakage flux.*

Flux can be divided into two main components

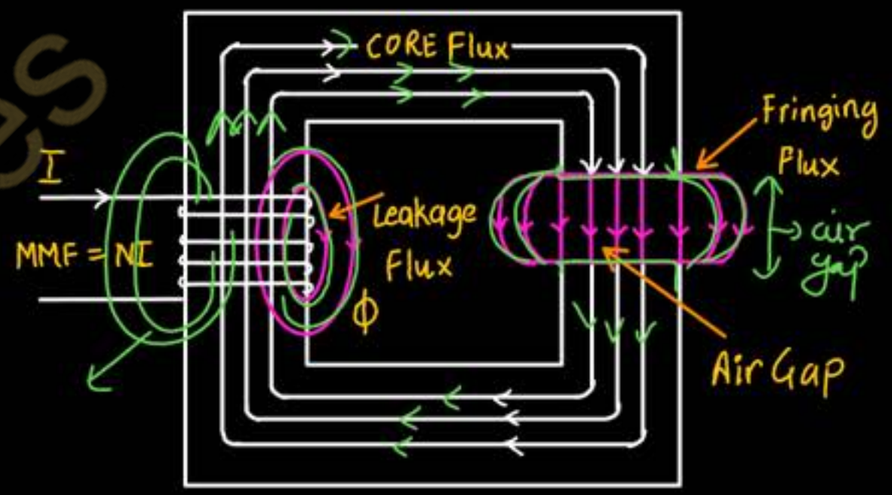
(i) magnetic flux throughout the magnetic circuit including air gap is known as ***useful flux*** **(ϕg)**

(ii) leakage flux *Link partially with magnetic circuit and complete its path through air.*

Leakage Flux = Total flux- useful flux

$$\phi_{leakage} = \phi_i - \phi_g$$

$$\text{Leakage Coefficient, } \lambda = \frac{\text{Total Magnetic Flux } (\phi i)}{\text{Useful Magnetic Flux } (\phi g)}$$



Magnetic Fringing:- When the magnetic field lines pass through an air gap, they tend to bulge out . It is because the magnetic field lines repel each other when passing through the air . This effect is known as *magnetic fringing.*

Due to magnetic fringing, the effective area of the air gap is increased and thus the magnetic flux density is decreased in the air gap. The longer the air gap, the higher is the fringing and *vice-versa.*

Gateway classes

Electromagnetic Induction

only for understanding

Faraday's law of Electromagnetic Induction

First law: Whenever a conductor is placed in a varying magnetic field, EMF induces and this emf is called an induced emf and if the conductor is a closed circuit than the induced current flows through it.

Second law: The magnitude of the induced EMF is equal to the rate of change of flux linkages.

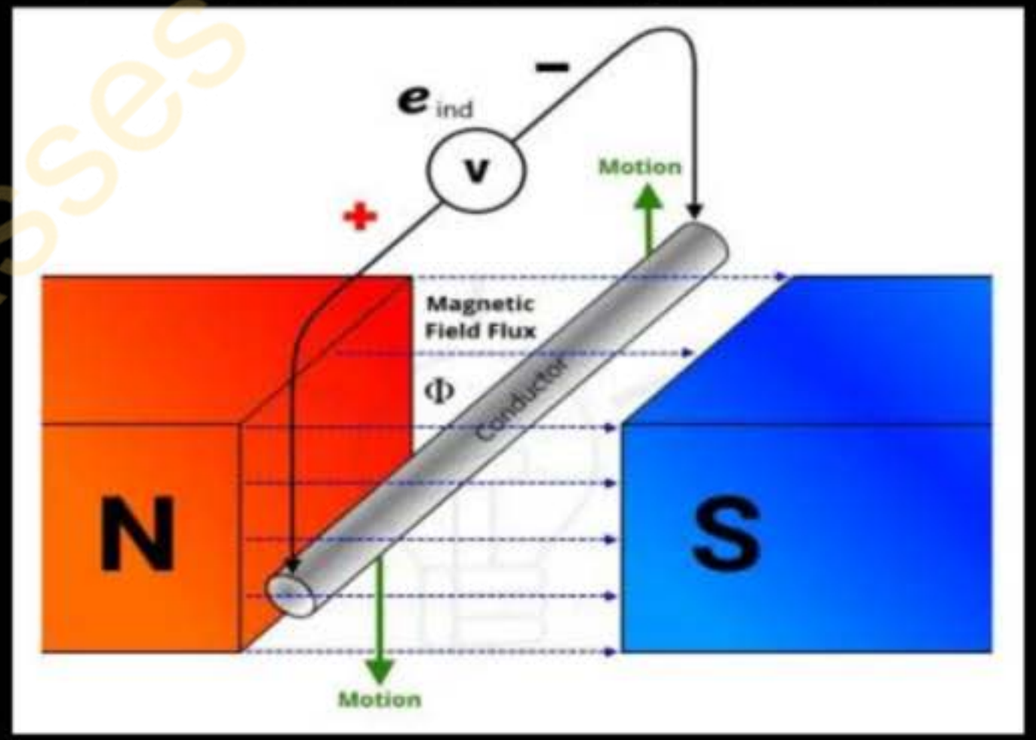
$e = N \frac{d\phi}{dt}$

Based on his experiments we now have Faraday's law of electromagnetic induction according to which the amount of voltage induced in a coil is proportional to the number of turns and the changing magnetic field of the coil.

So now, the induced voltage is as follows:

$$e = N \times d\Phi / dt$$

Faraday's Laws of Electromagnetic Induction



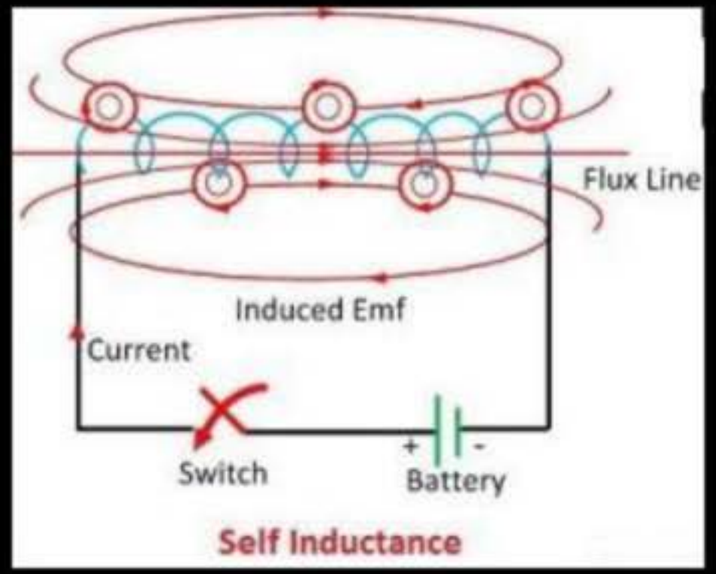
Gateway Classes

Self and mutual Inductance

Self-inductance :- It is defined as the property of the coil due to which it opposes the change of current flowing through it. Inductance is attained by a coil due to the self-induced emf produced in the coil itself by changing the current flowing through it.

If the current in the coil is increasing, the self-induced emf produced in the coil will oppose the rise of current, that means the direction of the induced emf is opposite to the applied voltage

$$e = L\frac{di}{dt}$$

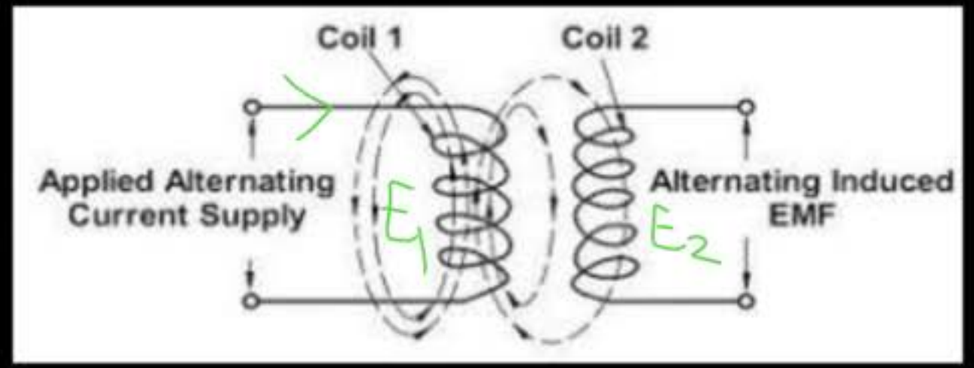


Self Inductance

Mutual Inductance between the two coils is defined as the property of the coil due to which it opposes the change of current in the other coil, or you can say in the neighboring coil.

When the current in the neighboring coil changes, the flux sets up in the coil and because of this, changing flux emf is induced in the coil called Mutually Induced emf and the phenomenon is known as **Mutual Inductance**.

$$e_m = M\frac{dI_1}{dt}$$



Gateway classes

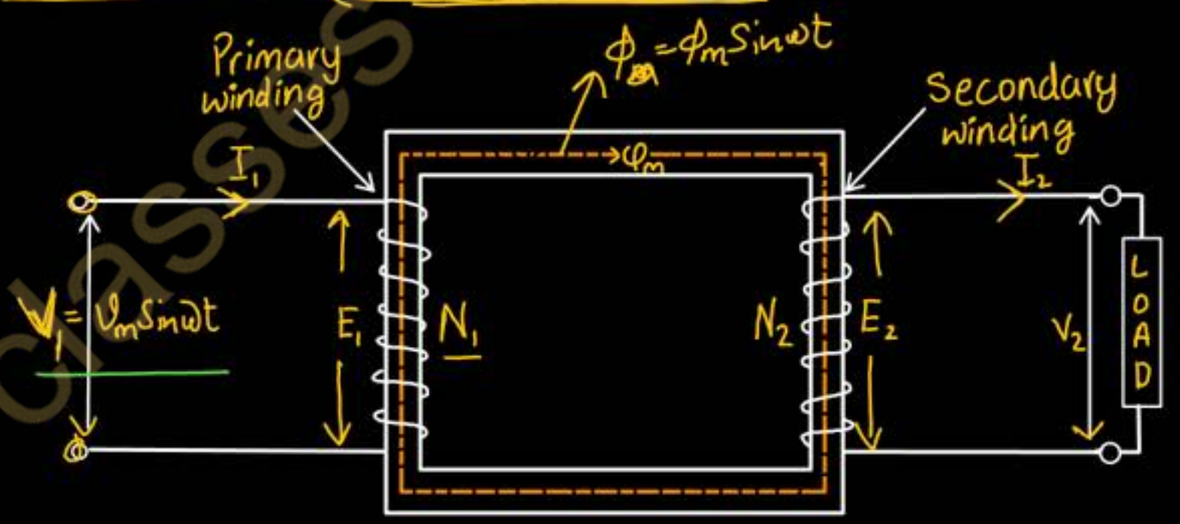
Transformer

$\downarrow V_1 \; I_1 \uparrow \; P = P \; I_2 \downarrow \; V_2 \uparrow$

A *transformer* is a static electrical machine which is used for either increasing or decreasing the voltage level of the AC supply with a corresponding decrease or increase in the current at constant frequency.

Working Principle of Transformer

- The **basic principle behind working of a transformer** is the phenomenon of mutual induction between two windings linked by common magnetic flux.
- Transformer consists of two inductive coils; primary winding and secondary winding.
- When, primary winding is connected to a source of alternating voltage, alternating magnetic flux is produced around the winding.



- The core provides magnetic path for the flux, to get linked with the secondary winding.
- As the flux produced is alternating, EMF gets induced in the secondary winding according to Faraday's law of electromagnetic induction.
- This emf is called 'mutually induced emf', and the frequency of mutually induced emf is same as that of supplied emf.
- If the secondary winding is closed circuit, then mutually induced current flows through it, and hence the electrical energy is transferred from one circuit (primary) to another circuit (secondary).

Gateway classes

Concept

Transformer (Turns Ratio):

$E_1 = -N_1 \, d\phi_m / dt$

and

$E_2 = -N_2 \, d\phi_m / dt$

$$K = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

Therefore

$E_2 / E_1 = N_2 / N_1$

2 marks

Ques: Can Transformer work on DC

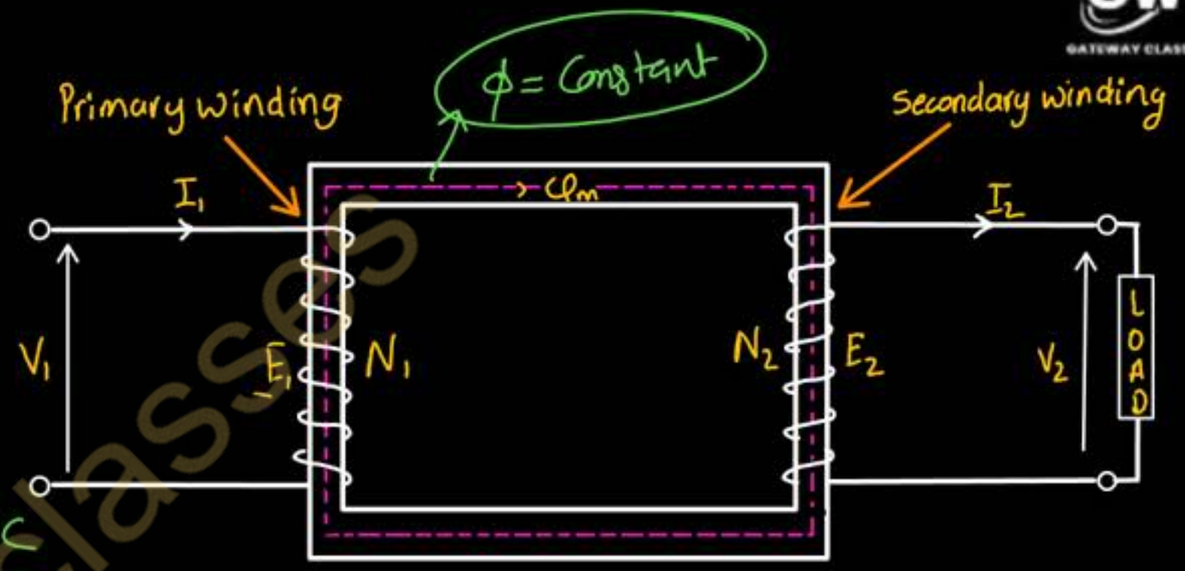
Ans.: No,

ϕ = Constant

$E = 0$

$I = 0$

D.C



Gateway classes

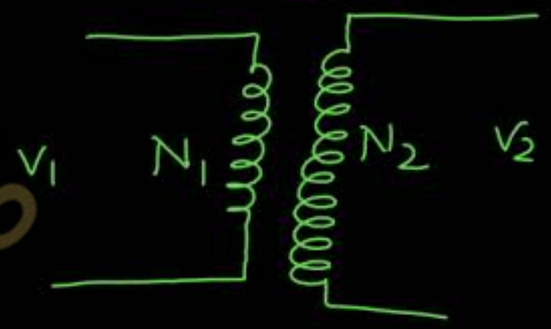
Types Of Transformers

Transformers can be classified on different basis, like types of construction, types of cooling etc.

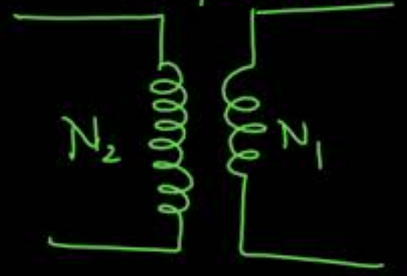
On the basis of their purpose

- **Step down transformer:-** A step-down transformer is used to decrease the voltage of an electrical current. It does this by using a larger number of turns in the secondary coil. $V_2 > V_1$, $N_2 > N_1$
- **Step up transformer:-** A step-up transformer is used to increase the voltage of an electrical current. The step-up transformer accomplishes this by using a larger number of turns in the primary coil. $V_2 < V_1$, $N_2 < N_1$

Step up



step down



On the basis of construction

- Core type transformer
- Shell type transformer

On the basis of type of supply

- Single phase transformer
- Three phase transformer

Gateway classes

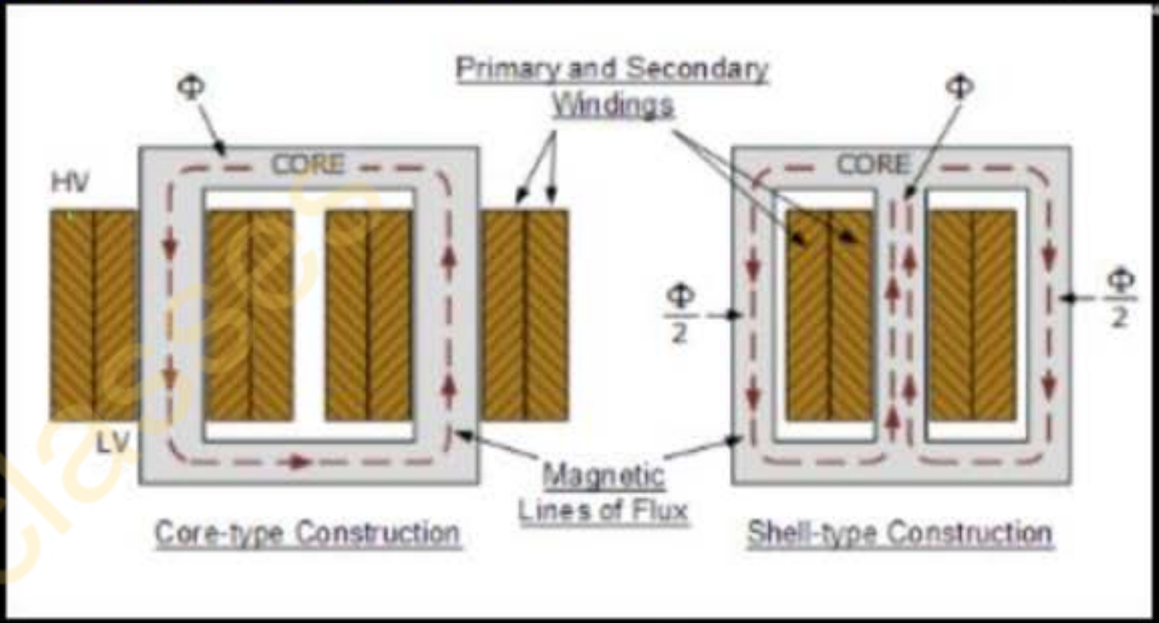
Classification of Transformer based on Construction

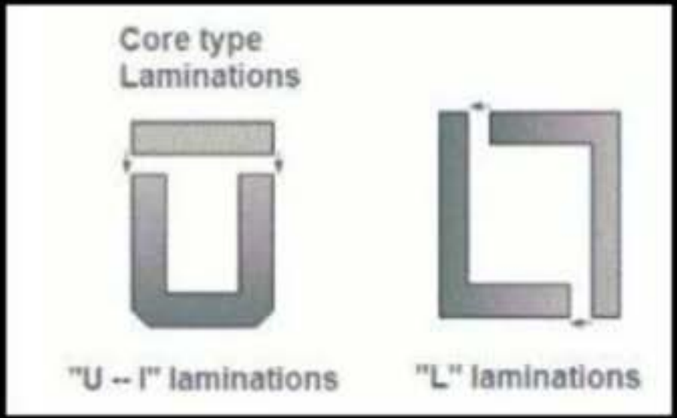
1. Core type transformer: In core type transformer winding surrounds a considerable part of core. Hence it is known as core type transformer.

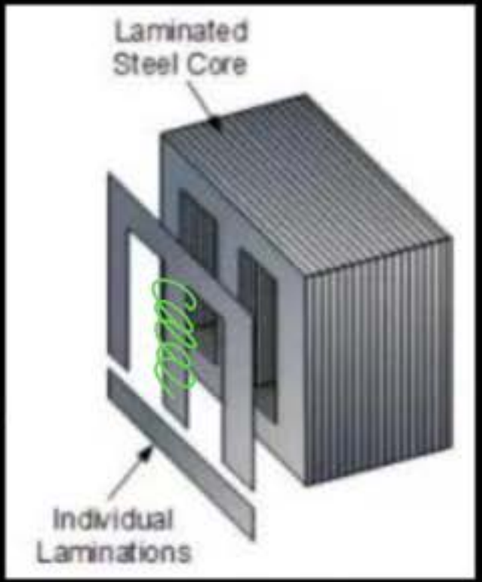
The core-type transformer is widely used for power transmission and distribution applications.

2. Shell-type transformer: In shell type transformer core surrounds a considerable part of winding. Hence it is known as shell type transformer.

Shell type transformers are mainly used for low voltage applications, often in low voltage power and electronic circuits. They help optimize circuit costs due to their square or rectangular cross-sectional cores, which are less expensive.







Gateway classes

E.M.F. Equation of Transformer

Imp

Let flux

$$\phi = \phi_m \sin\omega t$$

Emf induced on Primary side

$$E_1 = -N_1 \frac{d\phi}{dt}$$

$$E_1 = -N_1 \frac{d\,\phi_m \sin\omega t}{dt}$$

$$E_1 = -N_1 \phi_m \omega \cos\omega t$$

$$E_1 = \underbrace{N_1 \phi_m \omega}\sin\left(\omega t - \frac{\pi}{2}\right)$$

Compare it with standard Equation

$$E = E_m \sin\omega t$$

So

$$E_m = N_1 \phi_m \omega$$

$$E_{1rms} = \frac{E_m}{\sqrt{2}}$$

$$= \frac{N_1 \phi_m \omega}{\sqrt{2}}$$

$$E_1 = \frac{N_1 \phi_m 2\pi f}{\sqrt{2}}$$

$$\boxed{E_1 = 4.44\, N_1 \phi_m f} \text{ volt}$$

Similarly

$$\boxed{E_2 = 4.44\, N_2 \phi_m f}$$

Gateway Classes

2 marks Imp

Ques : Why Rating of Transformer is in KVA?

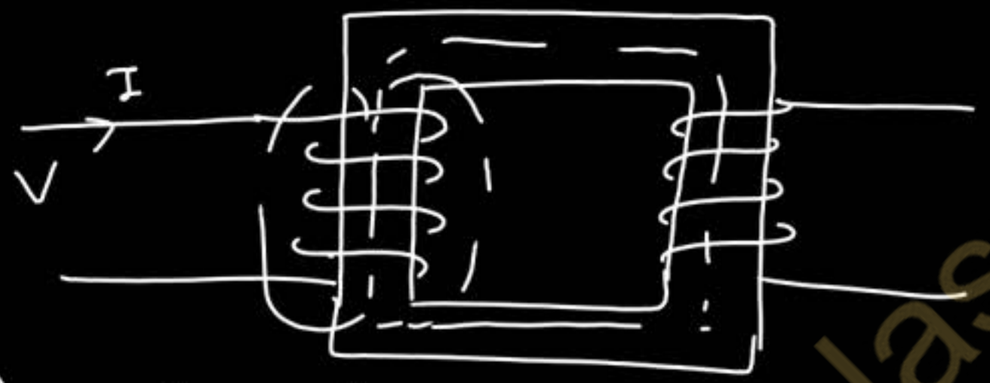
KW

Ans:- Iron loss on voltage and copper loss of a transformer depends on current. Hence, total transformer heat loss depends on volt-ampere (VA) and independent of phase angle among voltage and current i.e., it is independent of power factor. That is why the rating of the transformer is in kVA and not in kW.

Losses
- → Core loss $\propto$ Voltage
- → Copper loss $\propto$ I

Gateway classes

Ideal and Practical Transformer



In ideal transformer

leakage flux = 0

I^2R loss = 0

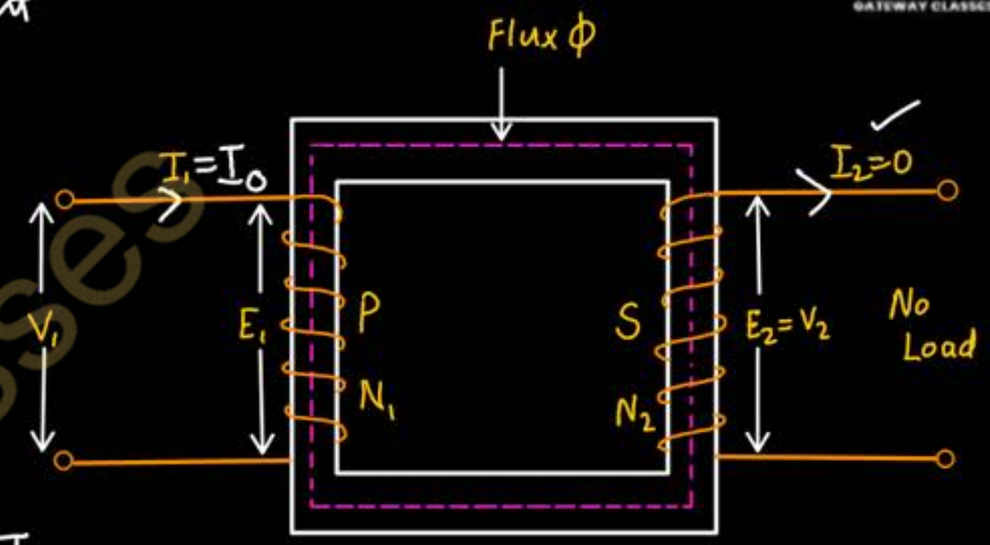
Gateway classes

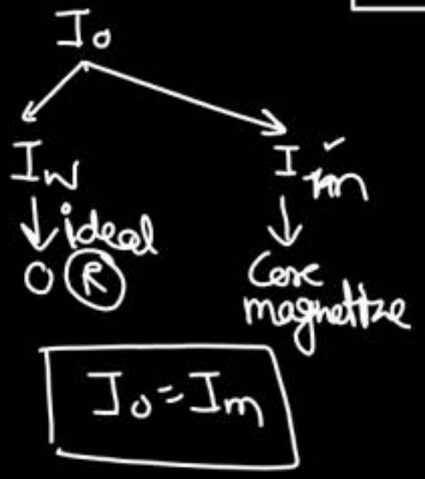
2 marks

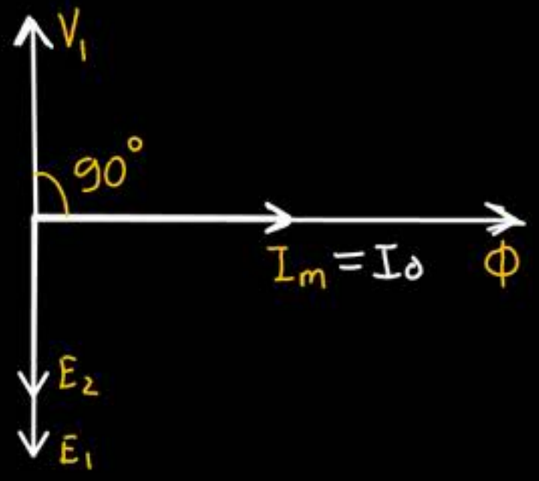
Ideal Transformer on no load

(5-8% of f.L. Current

- The primary draws a very small current I_1 which is just necessary to produce flux in the core. As it is magnetizing the core, it is called **magnetizing current denoted as I_m.**
- As the transformer is ideal, the winding ~~reactance~~ Resistance is zero and it is purely inductive in nature.
- The magnetizing current is very small and lags V_1 by 90° as the winding is purely inductive. This I_m produces an alternating flux Φ which is in phase with I_m.
- The flux links with both the winding producing the induced emf E1 and E2, in the primary and secondary windings respectively.
- According to Lenz's law, the induced emf opposes the cause producing it which is supply voltage V1. Hence E1 is in antiphase with V1 but equal in magnitude.
- The induced emf E2 also opposes V1 hence in antiphase with V1 but its magnitude depends on N2. Thus E1 and E2 are in phase.







Gateway classes

Ideal Transformer on no load

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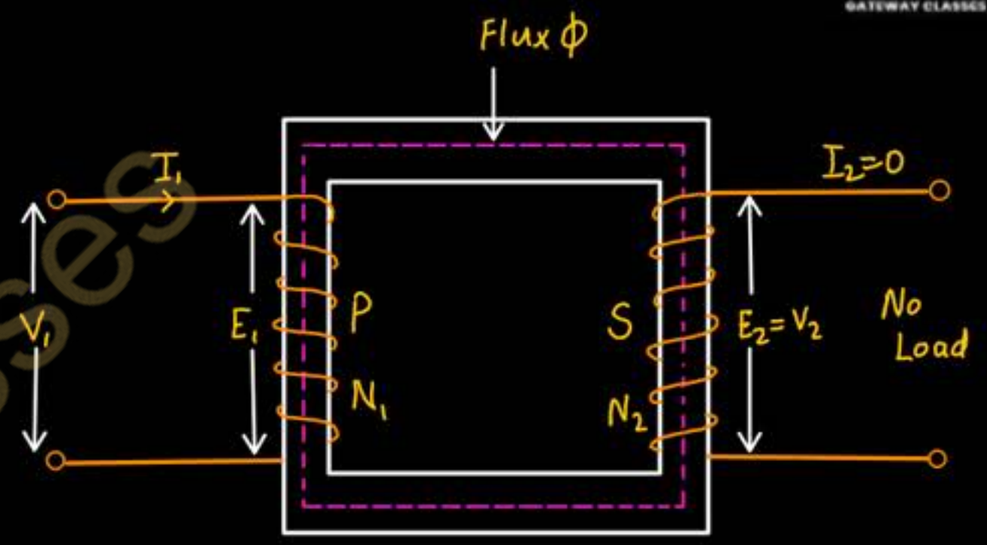
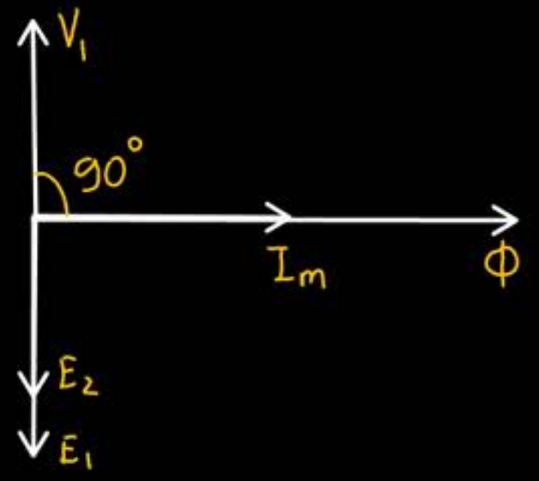
- The induced emf E2 also opposes V1 hence in antiphase with V1 but its magnitude depends on N2. Thus E1 and E2 are in phase.





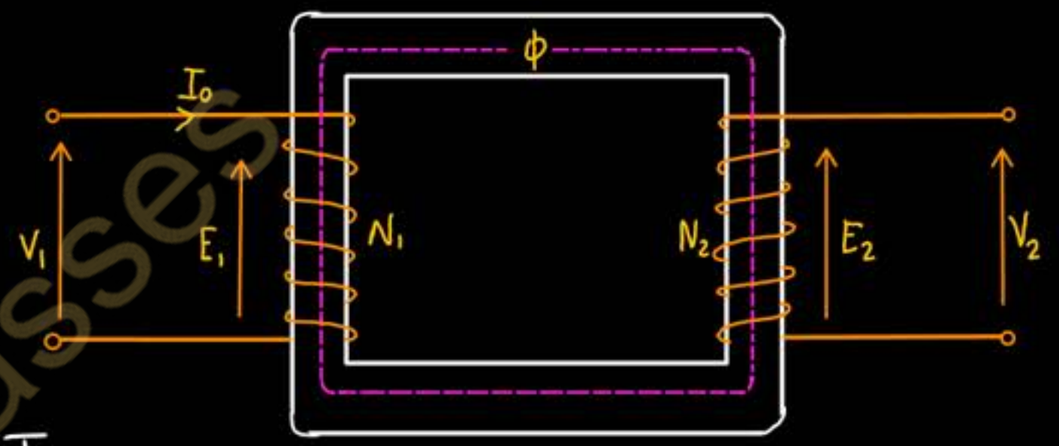
Gateway classes

Practical Transformer on no load

The no-load current consists of two components:

Reactive or magnetizing component Im:- It produces flux in the core

Active or power component I_w:- It supplies small amount of primary core loss.



Hence the primary no load current I_0 is not 90° behind the applied voltage V_1 but lags it by an angle $\Phi_0 < 90°$ as shown in the phasor diagram.

Working Component $I_w = I_0 \cos\phi_0$

Magnetizing Component $I_\mu = I_0 \sin\phi_0$

No Load Current $I_0 = \sqrt{I_w^2 + I_\mu^2}$

Power factor $\cos\phi_0 = \frac{I_w}{I_0}$

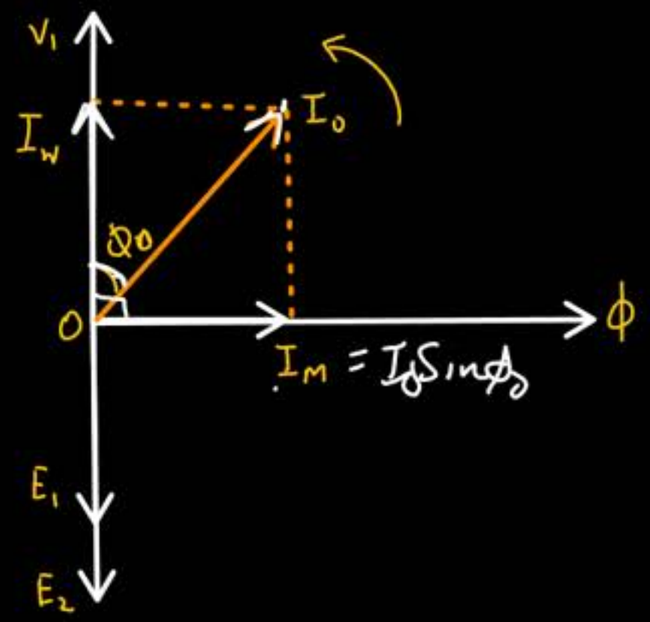
No Load power I/P $P_0 = V_1 I_0 \cos\phi_0$

I_0

I_w — Core loss — in phase with V

I_μ — magnetize — ϕ setup E_0

I_0



Gateway classes

Practical Transformer on no load

Ex.1 A 25 KVA , 3300/ 230V , 50 Hz , 1 phase transformer draws no load current of 15 A when excited on ~~load~~ Low voltage side and consumes 350 watts. Calculate two components of current . Aktu – 2003-04

Sol^n

3300 V / 230 V

$V_2 = 3300$, $V_1 = 230V$

$I_0 = 15 A$

$P = 350 W$

$P = V_1 I_0 \cos\phi_0$

$\cos\phi_0 = \frac{P}{V_1 I_0} = \frac{350}{230 \times 15}$

$\cos\phi_0 = 0.101$

$\phi_0 = 84.17$

$\sin\phi_0 = 0.99$

$I_w = I_0 \cos\phi_0 = 15 \times 0.101$

$= 1.51$ Amp

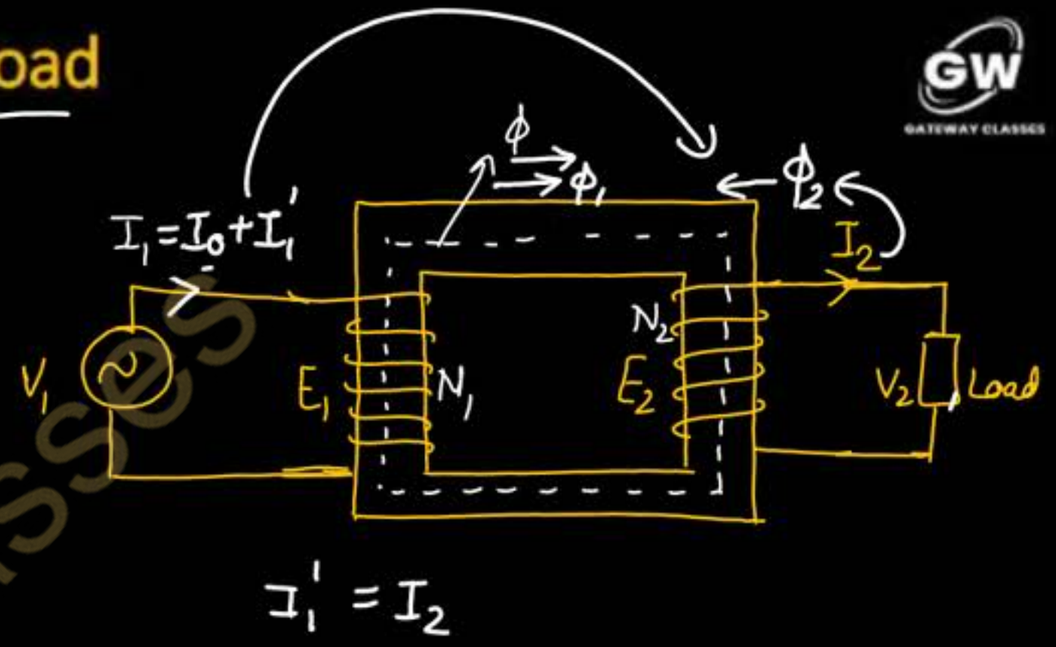
$I_m = I_0 \sin\phi_0$

$= 15 \times 0.99$

$I_m = 14.85$ A

Gateway classes

Transformer on Load



- When the secondary is kept open, it draws the no-load current from the main supply. The no-load current induces the mmf N_0I_0 and this force set up the flux Φ in the

- When the load is connected to the secondary of the transformer, I_2 current flows through their secondary winding. The secondary current induces the mmf N_2I_2 on the secondary winding of the transformer. This force set up the flux ϕ_2 in the transformer core. The flux ϕ_2 opposes the flux ϕ, according to **Lenz's law**.

- flux ϕ_2 opposes the flux ϕ, the resultant flux of the transformer decreases and this flux reduces the induced EMF E_1. Thus, the strength of V_1 is more than E_1 and an additional primary current I'_1 drawn from the mains.
- The additional current is used for restoring the original value of the flux in the core of the transformer so that $V_1 = E_1$. The primary current *I'_1 is in phase opposition with the secondary current I_2.*

- The additional current I'_1 induces the mmf $N_1I'_1$. And this force set up the flux ϕ'_1. The direction of the flux is the same as that of the ϕ and it cancels the flux ϕ_2 which induces because of the MMF N_2I_2

Gateway classes

Transformer on Load

$I_1 = I_0 + I_1'$

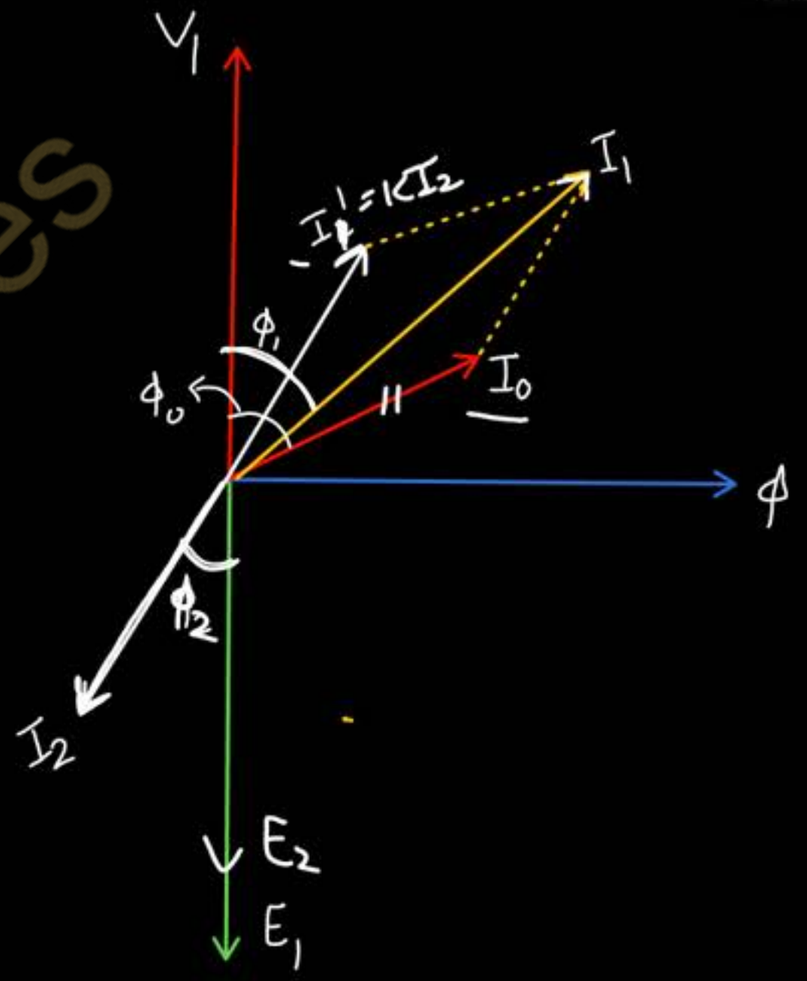
I_2

- Now, $\frac{N_1}{I_1'} = \frac{N_2}{I_2}$ Therefore,

$$I'_1 = \left(\frac{N_2}{N_1}\right) I_2 = KI_2$$

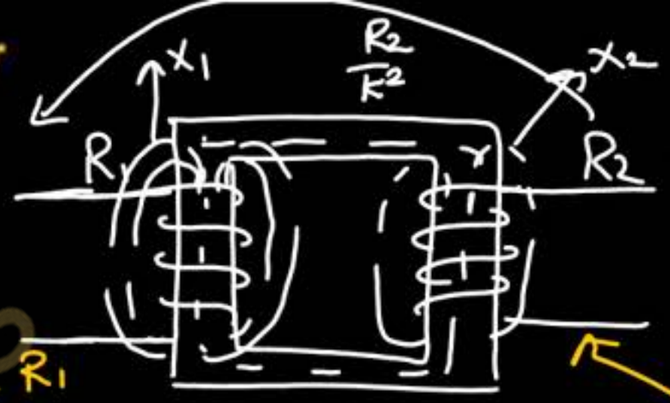
- The phase difference between V_1 and I_1 gives the power factor angle φ1 of the primary side of the transformer.
- The power factor of the secondary side depends upon the type of load connected to the transformer.
- The total primary current I1 is the vector sum of the currents I_0 and I_1'.

$$\boxed{\bar{I}_1 = \bar{I}_0 + \bar{I}'_1}$$



Gateway classes

Equivalent circuit of Transformer



winding resistance, R_1, R_2
leakage flux, X_1, X_2

Concept of Leakage reactance:-

Primary: R_1, X_1

Secondary: R_2, X_2

Concept of equivalent resistance:- referred to primary and secondary

Total loss = Losses on Primary + Losses on secondary

$$= I_1^2 R_1 + I_2^2 R_2$$

$$= I_1^2 \left[R_1 + R_2 \left(\frac{I_2}{I_1} \right)^2 \right]$$

$$= I_1^2 \left[\underbrace{R_1 + \frac{R_2}{K^2}} \right] = I_1^2 R_{01}$$

$$\boxed{R_{01} = R_1 + \frac{R_2}{K^2}}$$

$$= I_2^2 \left(R_2 + R_1 \left(\frac{I_1}{I_2} \right)^2 \right)$$

$$= I_2^2 (R_2 + R_1 K^2)$$

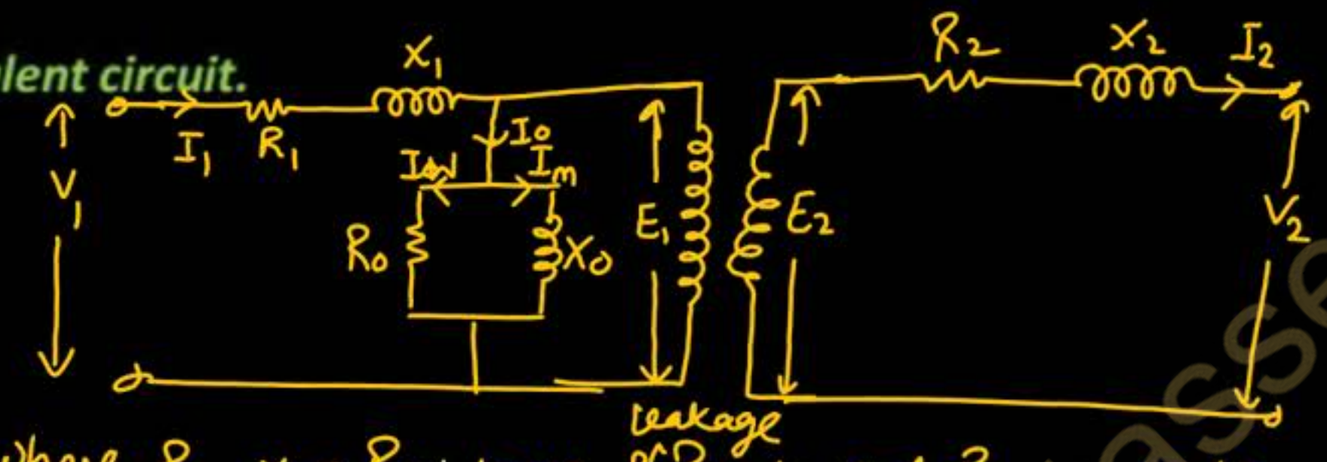
$$= I_2^2 R_{02}$$

$$\boxed{R_{02} = R_2 + K^2 R_1}$$

Gateway classes

Equivalent circuit of Transformer

Exact Equivalent circuit.



where R_1, X_1 = Resistance & Reactance of Primary side

R_2, X_2 = " " " Secondary "

E_1 = Induced Emf on Primary side

E_2 = " " " Secondary "

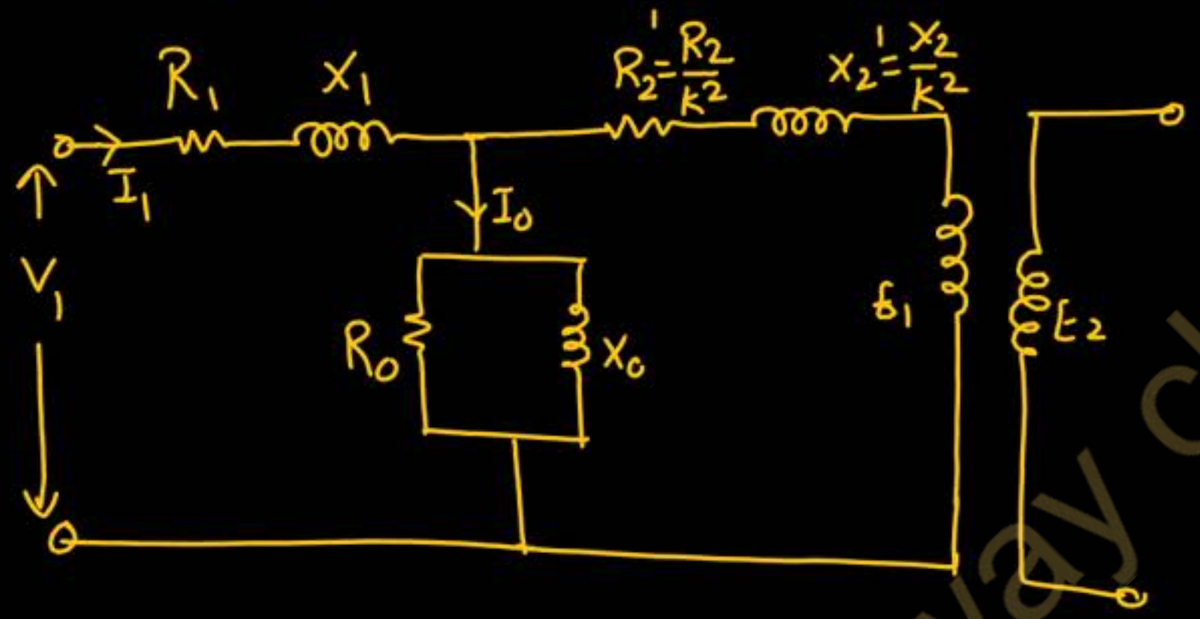
R_0 → Core loss Component

X_0 → magnetizing Component

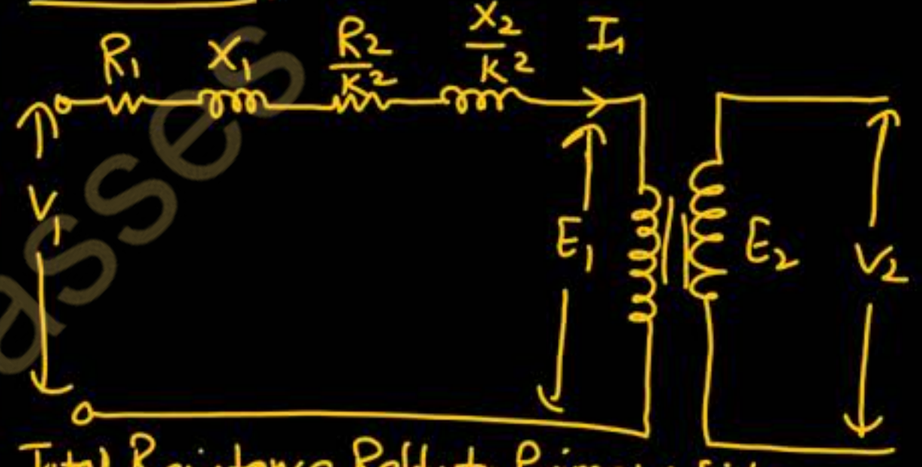
Gateway classes

Equivalent circuit of Transformer

Exact Equivalent circuit referred to Primary



Approximate Equivalent Circuit Ref to Primary



Total Resistance Reff. to Primary side

$$R_{01} = R_1 + \frac{R_2}{K^2}$$

Reactance

$$X_{01} = X_1 + X_2/K^2$$

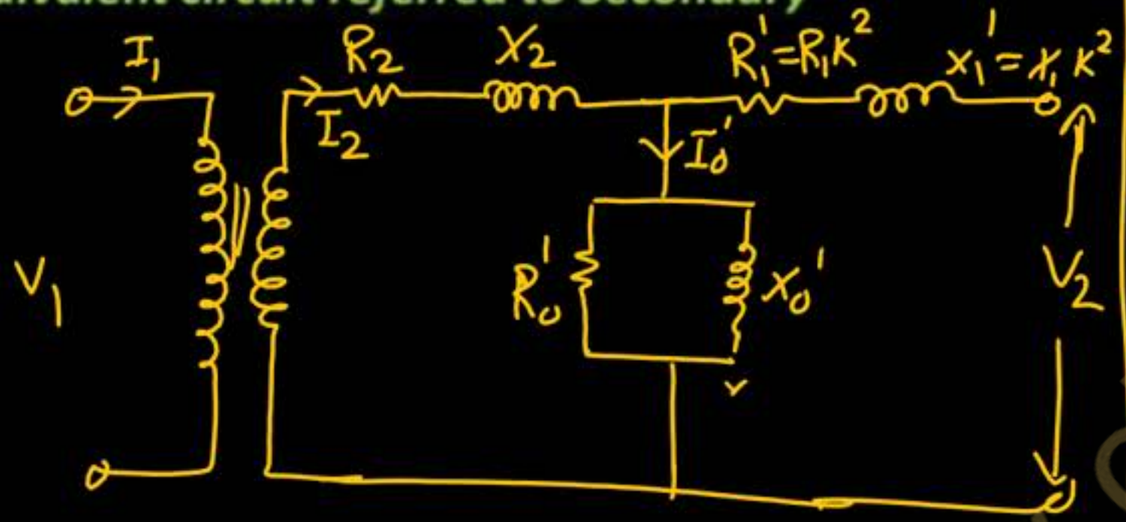
Total Impedance Ref. to Primary

$$Z_{01} = R_{01} + jX_{01} = \sqrt{R_{01}^2 + X_{01}^2}$$

Gateway classes

Equivalent circuit of Transformer

Exact Equivalent circuit referred to Secondary



App. Equivalent Circuit Ref. to Secondary

Total Resistance, Reactance & Impedance Ref. to Sec.

$$R_{02} = R_2 + K^2 R_1$$

$$X_{02} = X_2 + K^2 X_1$$

$$Z_{02} = R_{02} + jX_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$

Gateway classes

Numerical

Imp

V_1 V_2

Ex-1 A 20kVA, 2000V/200V, single-phase, 50 Hz transformer has a primary resistance of 1.5 Ω and reactance of 2 Ω. The secondary resistance and reactance are 0.015 Ω and 0.02 Ω respectively. The no load current of transformer is 1A at 0.2 power factor. Determine: (i) Equivalent resistance, reactance referred to primary & Secondary (ii) Total copper loss AKTU 2023-24, 2021-22

Sol^n 20KVA, $I_0 = 1A$, $\cos\phi_0 = 0.2$

$$K = \frac{V_2}{V_1} = \frac{200}{2000} = \frac{1}{10}$$

$R_1 = 1.5\Omega$, $X_1 = 2\Omega$

$R_2 = 0.015\Omega$, $X_2 = 0.02\Omega$

(i) Equivalent Resistance & Reactance Reff to Primary

$$R_{01} = R_1 + \frac{R_2}{K^2} = 1.5 + \frac{0.015}{(1/10)^2} = 3\Omega$$

$$X_{01} = X_1 + \frac{X_2}{K^2} = 2 + \frac{0.02}{(1/10)^2} = 4\Omega$$

$$P_{cu} = I_1^2 R_{01}$$

$$I_1 = \frac{P}{V_1} = \frac{20\times10^3}{2000} = 10A$$

$$P_{cu} = I_1^2 R_{01}$$

$$= 10^2 \times 3$$

$$\boxed{P_{cu} = 300 \text{ watt}}$$

(ii) Total Resistance & Reactance Ref to Secondary.

$$R_{02} = R_2 + K^2 R_1 = 0.015 + \frac{1.5}{100}$$

$$= 0.030\Omega$$

$$X_{02} = X_2 + K^2 X_1 = 0.02 + \frac{2}{100} = 0.040\Omega$$

$$I_2 = \frac{P}{V_2} = \frac{20\times10^3}{200} = 100A$$

$$P_{cu} = I_2^2 R_{02}$$

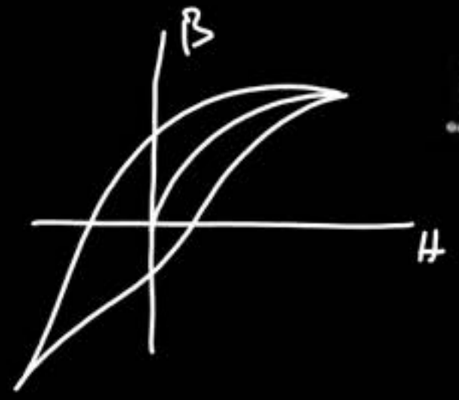
$$= 100\times100\times0.030$$

$$\boxed{P_{cu} = 300 \text{ watt}}$$

Gateway classes

Losses in a Transformer

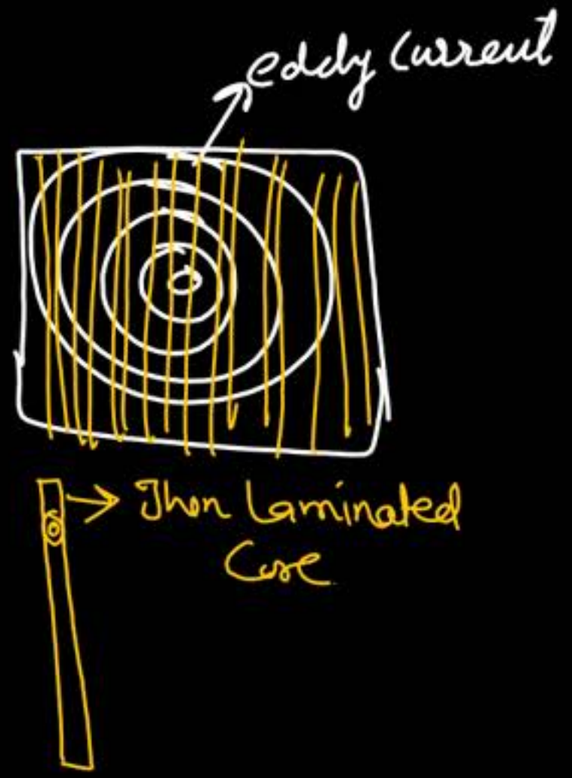
Core Loss/ Iron Loss/ Fixed Loss/ Constant Loss



Hysteresis Loss:- When a magnetic material is subjected to cycle of magnetization , a power loss occurs due to molecular friction in the material Therefore, energy is required in the material to overcome this opposition. This loss being in the form of heat and is termed as *hysteresis loss.*

$$\text{Area of hysteresis loop} \propto B_{max}^{1.6}$$

Hysteresis power loss, $P_h = \eta B_{max}^{1.6} f V$ ***Watts***



Eddy Current Loss:- When a magnetic material is subjected to a changing magnetic field, a voltage is induced in the material according to Faraday's law of electromagnetic induction. Since the material is conducting, the induced voltage circulates currents within the body of the magnetic material. These circulating currents are known as *eddy currents*. These eddy currents causes I^2R loss in the material, known as *eddy current loss*.

Eddy current power loss, $P_e = K_e B^2_{max} f^2 t^2 V$ ***Watts***

Where *K_e = Eddy current coefficient, B_{max} = Maximum flux density, f = frequency of magnetization or flux, t = thickness of lamination, and V = Volume of magnetic material.*

Gateway classes

Methods of reducing Losses in a Transformer

Copper Loss/ Variable Loss/ I^2R Loss

- **Copper Loss:-** The losses are occurs due the resistance of primary and secondary winding. These losses are depends upon the magnitude of current in winding.

$$P_{cu} \propto I^2$$

$$P_{cu} = I^2R$$

- *Thus for a Transformer Total Losses = Iron Loss + Copper Loss = $P_i + P_{cu}$*

Methods of reducing Losses in a Transformer

Iron Loss:- (i) Choice of Material for Iron Core (steel of high silicon content)

(ii) Use of Thin Laminated Core

Copper Loss/ Variable Loss/ I^2R Loss

(i) Reduce resistance of winding (use of thick wire)

Gateway classes

Efficiency of Transformer

- **Efficiency of Transformer:-**

$$\eta = \frac{\text{output}}{\text{Input}} = \frac{\text{output}}{\text{output} + \text{losses}}$$

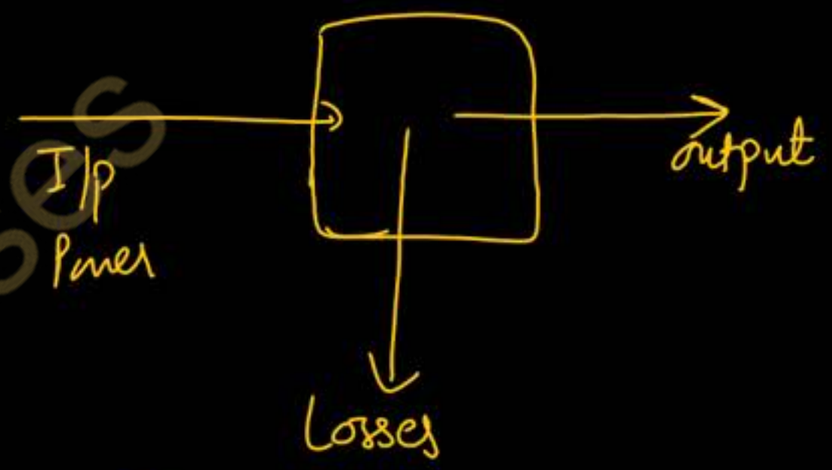
$$\boxed{\%\eta_{f.L} = \frac{V_2 I_2 \cos\phi}{V_2 I_2 \cos\phi + P_{i^o} + P_{cu}} \times 100}$$

Efficiency at fraction of Load

Let at half load $x = \frac{1}{2} = 0.5$

or At any load of fraction x

$$\eta_x = \frac{x V_2 I_2 \cos\phi}{x V_2 I_2 \cos\phi + P_{i^o} + x^2 P_{cu}}$$



Input − losses = output

Input = output + losses

Gateway classes

Efficiency of Transformer

V. Imp

Condition of Maximum Efficiency of a Transformer:-

$$\eta = \frac{V_2 I_2 \cos\phi}{V_2 I_2 \cos\phi + P_{i0} + I_2^2 R_{02}}$$

for maximum efficiency

$$\frac{d\eta}{dI_2} = 0$$

$$\frac{d\eta}{dI_2} = \frac{(V_2 I_2 \cos\phi + P_{i0} + I_2^2 R_{02})(V_2 \cos\phi) - (V_2 I_2 \cos\phi)(V_2 \cos\phi + 0 + 2 I_2 R_{02})}{V_2 I_2 \cos\phi + P_{i0} + I_2^2 R_{02}} = 0$$

$$= V_2 \cos\phi \left[\cancel{V_2 I_2 \cos\phi} + P_{i0} + I_2^2 R_{02} - \cancel{V_2 I_2 \cos\phi} - 2 I_2^2 R_{02} \right) = 0$$

$$P_{i0} - I_2^2 R_{02} = 0$$

$$\boxed{P_{i0} = P_{cu}}$$

when iron Loss = Copper loss then the η will be maximum.

for any other load

$$\boxed{P_{i0} = x^2 P_{cu}}$$

Gateway classes

Efficiency of Transformer

Load Current at Maximum Efficiency :-

at max. η

$$I_{2_{\eta m}}^2 R_{o2} = P_i^o$$

$$I_{2_{m\eta}} = \sqrt{\frac{P_i^o}{R_{o2}}}$$

$$\frac{I_{2m\eta}}{I_{2f.L}} = \frac{1}{I_{2f.L}}\sqrt{\frac{P_i^o}{R_{o2}}}$$

$$\frac{I_{2m\eta}}{I_{2f.L}} = \sqrt{\frac{P_i^o}{P_{cu_{f.L}}}}$$

$$\boxed{I_{2m\eta} = I_{2fL}\sqrt{\frac{P_i^o}{P_{cu}}}}$$

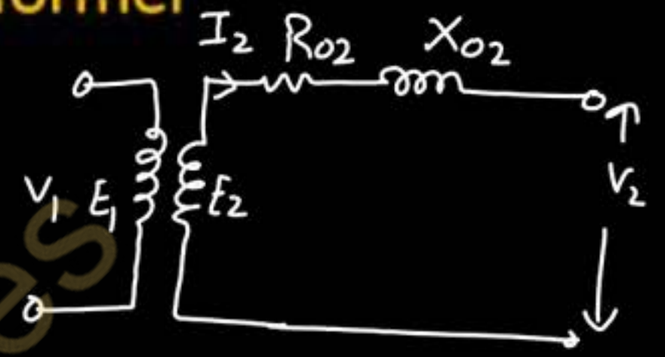
KVA at Maximum Efficiency:-

$$(KVA)_{\eta_m} = I_{2m\eta} \cdot V_2 = I_{2fL} \cdot V_2 \sqrt{\frac{P_i^o}{P_{cu}}} = (KVA)_{f.L}\sqrt{\frac{P_i^o}{P_{cu}}}$$

$$\boxed{(KVA)_{m\eta} = (KVA)_{f.L}\sqrt{\frac{P_i^o}{P_{cu}}}}$$

Gateway classes

Voltage Regulation of a Transformer



Approximate Voltage drop of transformer:-

for $E_2 = I_2 R_{02} + jI_2 X_{02} + V_2$

$$E_2 = I_2(R_{02} + jX_{02}) + V_2 \quad \text{—(1)}$$

Let OC = radius = E_2

$$E_2 - V_2 = OD - OA = Af \cong AD$$

AD = Total voltage drop = $AC + CD$

$$AC = I_2 R_{02} \cos\phi_2$$

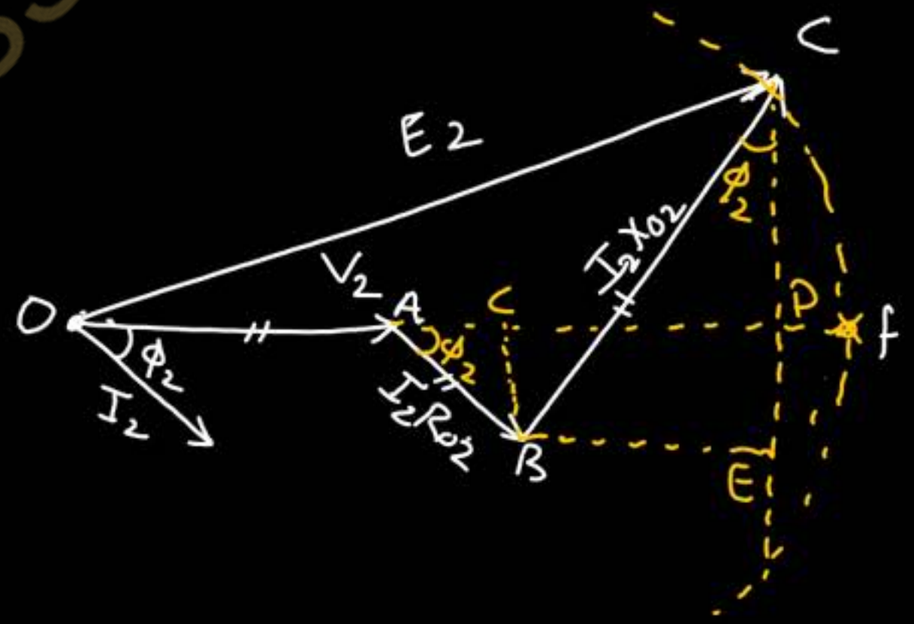
$$CD = BE = I_2 X_{02} \sin\phi_2$$

Total voltage drop (Lagg) $= I_2 R_{02} \cos\phi_2 + I_2 X_{02} \sin\phi_2$

$$= I_2(R_{02}\cos\phi_2 + X_{02}\sin\phi_2)$$

for Leading Power fact

$$= I_2(R_{02}\cos\phi_2 - X_{02}\sin\phi_2)$$



Gateway classes

Voltage Regulation of a Transformer

- Voltage regulation is the measure of how well a transformer can maintain a constant secondary voltage under varying load conditions, as the output secondary voltage may not be what we expect.
- Voltage regulation is defined as change in magnitude of the secondary terminal voltage, when full load (rated load) of specified power factor at rated voltage is reduced to no load, with primary voltage maintained constant expressed as the percentage of the rated terminal voltage.

$$\% \text{ Regulation} = \frac{E_2 - V_2}{V_2} \times 100$$

$$= \frac{I_2 R_{02} \cos\phi_2 \pm I_2 X_{02} \sin\phi_2}{V_2}$$

(+) → lagging

(−) → leading

Gateway classes

Numericals

Ex-1 A 40 KVA transformer has iron loss of 450W and full load copper loss of 850W. if the power factor of the load is 0.8 lagging, calculate: (i) Full load efficiency (ii) The load at which the maximum efficiency occurs and (iii) The maximum efficiency. 2021-22

Sol^n $P_i = 450W$, $P_{cu} = 850W$, $\cos\phi = 0.8$

$$\eta_{f.L} = \frac{V_2 I_2 \cos\phi}{V_2 I_2 \cos\phi + P_i + P_{cu}} = \frac{40\times10^3 \times 0.8}{(40\times10^3\times0.8) + 450 + 850} \times 100$$

$$= \frac{32000}{33300} \times 100$$

$$\% \eta_{f.L} = 96.09\%$$

(ii) $$(KVA)_{m\eta} = (KVA)_{f.L}\sqrt{\frac{P_i}{P_{cu}}} = 40\times10^3\sqrt{\frac{450}{850}}$$

$$= 29.10 \text{ KVA}$$

(iii) $$\eta_m = \frac{29.10\times10^3 \times 0.8}{29.10\times10^3\times0.8 + 2P_i}$$

$P_{cu} = P_i$

$$= \frac{23283.42}{23283.42 + 2\times450}$$

$$\% \eta_m = \frac{23283.42}{24183.42} \times 100$$

$$\boxed{\% \eta_m = 96.27\%}$$

Gateway classes

Numericals

V_2I_2

Ex-2 A 100 kVA, 1-φ transformer has iron loss of 600 W and a copper loss of 1.5 kW at full-load current. Calculate the efficiency at (i) full load and 0.8 pf (lagging), and (ii) half load and unity pf? AKTU-2023-24

$P_{i0} = 600\ W, \quad P_{cu} = 1500\ W$

$$\eta_{f.L} = \frac{100\times10^3\times0.8}{100\times10^3\times0.8 + 600 + 1500}$$

$$\boxed{\%\eta = 97.44\%}$$

(ii) $\cos\phi = 1, \quad x = \frac{1}{2}$

$$\eta_{H.L} = \frac{x V_2 I_2 \cos\phi}{x V_2 I_2 \cos\phi + P_{i0} + x^2 P_{cu}}$$

$$= \frac{\frac{1}{2}\times100\times1\times10^3}{\frac{1}{2}\times100\times10^3\times1 + 600 + \frac{1500}{4}}$$

$$= \frac{50000}{50000 + 600 + \frac{1500}{4}}$$

$$\boxed{\%\eta = 98.08\%}$$

Gateway classes

Numericals

Ex-3 A single phase 100KVA [illegible], 230V 50Hz transformer has 90% efficiency at 0.8 lagging power factor both at full load and also at half load. Determine iron and copper loss at full load for the transformer. 2019-20

at f.L load

$\eta = 90\%$

$$\eta = \frac{V_2 I_2 \cos\phi}{V_2 I_2 \cos\phi + P_i + P_{cu}}$$

$$0.9 = \frac{100 \times 10^3 \times 0.8}{100 \times 10^3 \times 0.8 + P_i + P_{cu}}$$

$$0.9 \times (80000 + P_i + P_{cu}) = 80000$$

$$0.9(P_i + P_{cu}) = 80000 - 72000$$

$$\boxed{P_i + P_{cu} = \frac{8000}{0.9}} \quad \text{—} (1)$$

=

at H.L $x = 0.5 = \frac{1}{2}$

$$\eta = \frac{x V_2 I_2 \cos\phi}{x V_2 I_2 \cos\phi + P_i + P_{cu} x^2}$$

$$0.9 = \frac{\frac{1}{2} \times 100 \times 10^3 \times 0.8}{\frac{1}{2} \times 100 \times 10^3 \times 0.8 + P_i + \frac{P_{cu}}{4}}$$

$$0.9 \times \left(40000 + P_i + \frac{P_{cu}}{4}\right) = 40000$$

$$\left(P_i + \frac{P_{cu}}{4}\right) 0.9 = 40000 - 36000$$

$$P_i + \frac{P_{cu}}{4} = \frac{4000}{0.9} = \text{—} (2)$$

Solve Eqn (1) & (2)

$P_i = 2962.96$ watt

$P_{cu} = 5925.92$ watt

Gateway classes

Numericals

Ex-4 The maximum efficiency of a 100 KVA, 1100/440 V, 50 Hz transformer is 96%, This occurs at 75% of full load at 0.8 p.f. lagging. Find the efficiency of transformer at 3.4 FL at 0.6 p.f. leading. 2018-19

$\eta_m = 0.96$, $\cos\phi = 0.8$

$(KVA)_{m\eta} = 0.75 \times KVA_{f.L}$

$= .75 \times 100$

$= 75\ KVA$

$$\eta_m = \frac{75\times10^3\times0.8}{75\times10^3\times0.8 + 2P_i^o} = 0.96$$

$$\frac{60000}{60000+2P_i^o} = 0.96$$

$$0.96(60000+2P_i^o) = 60000$$

$$0.96\times2P_i^o = 60000-57600$$

$$= 2400$$

$$P_i^o = \frac{2400}{2\times.96}$$

$$\boxed{P_i^o = 1250\,W}$$

$$P_i^o = x^2 P_{cu}$$

$$P_{cu} = \frac{P_i^o}{x^2} = \frac{1250}{(.75)^2}$$

$$= 2222.2\ W$$

$$\eta_{3.4FL} = \frac{x V_2 I_2 \cos\phi}{x V_2 I_2 \cos\phi + P_i + x^2 P_{cu}}$$

$$= \frac{3.4\times100\times10^3\times0.6}{3.4\times100\times10^3\times0.6 + 1250 + 2222.2\times(3.4)^2}$$

$$= \frac{204000}{(204000+1250+\cancel{25688.86})\,25688.86}$$

$$\boxed{\eta_{3.4FL} = \cancel{96.96\%}\ 88.33\%}$$

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Thank You

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- → Equivalent Circuit + Num
- → Efficiency - + Num
- → Em.f Equation
- → Working of Transformer

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